Chapter 1

Rotational-state cleanup

After the electrostatic lens the CeNTREX molecules are in $X(v{=}0)$, $J = 2$, and the state-preparation stage SPB transfers them to $J = 1$, the rotational level used in the interaction region [1]. Two cleanup stages bracket this transfer. CU1, before SPB, removes the molecules that are already in $J = 1$, which would otherwise reach the interaction region together with the transferred ones. CU2, after SPB, removes the population left in $J = 2$. Both stages pump the molecules optically through the $B\,^3\Pi_1$ state into other rotational levels, in the electric field that is kept on to preserve the spin. This chapter describes the two schemes and their optical-Bloch-equation (OBE) model (Secs. 1.1 and 1.2), the CU1 simulation and a first test of CU1 on the SPB setup (Sec. 1.3), and the optimisation of CU2 at its present operating field (Sec. 1.4).

1.1 Optical pumping in an electric field

A cleanup stage drives a transition from the addressed rotational level J of X to B. Each excitation is followed by spontaneous decay, and since B decays to several rotational levels, most decays end in a level that the light does not address: from B, $J' = 2$, $F' = 2$ only 37 % of the decays return to $J = 1$, and from $J' = 3$, $F' = 3$ or $F' = 4$ only 27 % return to $J = 2$ (Table 1.1). A molecule is therefore removed after one or two photons, provided that every sublevel of the addressed level is excited and none is left in a dark superposition. States are labelled $|J, F_1, F, m_F, P\rangle$, with the nuclear spins coupled as $\mathbf{J} + \mathbf{I}_1 = \mathbf{F}_1$ and $\mathbf{F}_1 + \mathbf{I}_2 = \mathbf{F}$; primed quantum numbers refer to B, and R(J) denotes $J \to J' = J + 1$. The level $J = 1$ has 12 sublevels ($F_1 = \frac{1}{2}$: $F = 0, 1$; $F_1 = \frac{3}{2}$: $F = 1, 2$), $J = 2$ has 20.

The field, 150 V/cm for CU1 and 200 V/cm for CU2, defines the quantisation axis z; m_F remains a good quantum number, while J, F_1, F and the parity of B are mixed. The hyperfine structure of each ground rotational level spans only about 2 MHz, but in B the Stark interaction splits the $|m'_F|$ sublevels of one hyperfine level by 10–25 MHz each, many natural linewidths ($\Gamma/2\pi = 1.61$ MHz). A single laser frequency then reaches only a few excited sublevels, and for a fixed polarization N_g ground sublevels coupled to $N_e < N_g$ excited ones leave at least $N_g - N_e$ superpositions dark. Both schemes use three tools against this:

1. **reach enough excited sublevels**, with the carrier placed between them, with phase-modulation sidebands, or with a second beam on a second hyperfine level;
2. **switch the polarization** between X ($\perp \mathbf{E}$) and Z ($\parallel \mathbf{E}$) at 0.5 MHz, so that a superposition dark for one is bright for the other;
3. **use field mixing**, which makes zero-field-forbidden transitions driveable. For example,

$|J=1, F_1=\frac{1}{2}, F=0\rangle$ cannot be excited to $F'=2$ at zero field, but at 150 V/cm it carries a 29 % admixture of $|F_1=\frac{3}{2}, F=2\rangle$, through which CU1 pumps it.

Table 1.1: Cleanup stages and the parameters of the model. The main pair of each beam defines the zero of its detuning and normalises its Rabi rate; $|m|$ is its matrix element in units of the X–B transition dipole moment (X polarization). Beam heights are full 4σ widths.

	CU1 ($J=1$)	CU2 ($J=2$)
field	150 V/cm (test: 156 V/cm)	200 V/cm
transition	R(1) $F_1'=5/2$, $F'=2$	R(2) $F_1'=7/2$, $F'=3$ (beam 0) and $F'=4$ (beam 1)
main pair, $\lvert m\rvert$	$\lvert 1,\frac{3}{2},2,1\rangle \to \lvert 2,\frac{5}{2},2,2,+\rangle$, 0.100	$\lvert 2,\frac{5}{2},3,2\rangle \to \lvert 3,\frac{7}{2},3,3,-\rangle$, 0.095; $\lvert 2,\frac{5}{2},3,0\rangle \to \lvert 3,\frac{7}{2},4,1,-\rangle$, 0.229
decays back to J	37 %	27 %
OBE levels	21: 12 ($J=1$), 5 ($F'=2$, $P'=+$), 4 compacted	56: 20 ($J=2$), 32 ($F'=3,4$, both P'), 4 compacted
power	80 mW (test: 50 mW)	80 mW ($F'=3$) + 24 mW ($F'=4$)
beam height	4.8 cm; present 3 cm; test 2 cm	2.4 cm; present 3 cm
phase modulation	none	17.5 MHz, $\beta=1.5$, both beams
AOM	–	$f_{\mathrm{AOM}}=200\pm15$ MHz
multipass	13 passes 3.5 mm apart, 3 mm wide, 10 % loss per pass (CU1 test: about 17)	
polarization	X/Z square-wave switching at 0.5 MHz (CU2 beams in antiphase)	
molecule	184 m/s through 5 cm (CU1 test: 145 m/s)	
distribution	lensed beam (25 kV): $\sigma_{v_x}=0.65$ m/s, $\sigma_y\approx 11$ mm	
acceptance	$\lvert v_x\rvert\le 1$ m/s, $\lvert y_0\rvert\le 8.5$ mm (detection laser)	

1.2 Model

States. The X and B Hamiltonians of `centrex-tlf` [2] (rotation with centrifugal distortion, hyperfine and nuclear spin–rotation interactions, Λ-doubling of B, Stark and Zeeman interactions) are diagonalised in the applied field, with a magnetic field of 1×10^{-3} G along z that lifts the degeneracy of the $\pm m_F$ pairs. Each eigenvector is labelled by the bare state with the largest overlap; at 150 and 200 V/cm this agrees with adiabatic continuation from zero field. The OBE system keeps all sublevels of the addressed ground level and all m_F' of the addressed excited hyperfine levels, with all coherences; for CU2 both Λ-doublet components are kept, because the field mixes them strongly (for CU1 the partner changes the cleanup by 6×10^{-6}). Every other rotational level of X reachable by decay is compacted into one level, which is exact for the cleanup because no light acts on it (Table 1.1).

Hamiltonian and decay. In the rotating-wave approximation, with $\hbar=1$,

$$H(t)=H_0-\sum_k \delta_k\Pi_k+\frac{1}{2}\sum_k\Big[\Omega_k(t)\sum_{p\in\{X,Z\}} g_{p,k}(t)\,\frac{M_k^{(p)}}{m_k}+\mathrm{h.c.}\Big], \tag{1.1}$$

where H_0 holds the field-dressed energies, Π_k projects onto the excited states of beam k, $M_k^{(p)}$ is the dipole matrix between the field-dressed states for polarization p, normalised by the main-pair element m_k, and $g_{p,k}(t)$ is the polarization gate. Whether a pair is coupled

is decided by this mixed-state matrix element, not by the zero-field selection rules. The detuning δ_k is measured from the main resonance of beam k. Phase modulation enters as $\Omega_k \to \Omega_k e^{\mathrm{i}\beta \sin \omega_\mathrm{m} t}$; for CU2 both beams are written in the rotating frame of beam 0, so the couplings of beam 1 carry $e^{2\pi \mathrm{i} f_\mathrm{AOM} t}$. The excited states decay at $\Gamma = 1/\tau = 1.01 \times 10^7\,\mathrm{s}^{-1}$ ($\tau = 99\,\mathrm{ns}$), with branching ratios computed between the dressed states into every X level, and

$$\dot{\rho} = -\mathrm{i}[H, \rho] + \sum_{e,g} \Big(C_{eg} \rho C_{eg}^\dagger - \tfrac{1}{2} \{ C_{eg}^\dagger C_{eg}, \rho \} \Big), \qquad C_{eg} = \sqrt{\Gamma b_{e \to g}}\, |g\rangle\langle e|. \tag{1.2}$$

With one collapse operator per pair, decay transfers populations but no ground-state coherences. The system is closed, $\mathrm{Tr}\,\rho = 1$ to 10^{-14}. Decay to $X(v \geq 1)$ is not modelled, which can only underestimate the cleanup.

Laser and trajectory. The light crosses the molecular beam 13 times in a two-prism multipass. The passes are Gaussian, with $\sigma_s = 0.75\,\mathrm{mm}$ along the beam and σ_y equal to a quarter of the beam height across it. The pass crossed p-th in optical order carries the power fraction $(1 - L)^{p-1}$, with L the loss per pass; the passes add incoherently in intensity, and $|\Omega_k| \propto \sqrt{I_k}$. In the laser frame z is along the field (vertical in the laboratory) and x is horizontal along the molecular beam; the light propagates horizontally across it. Consecutive passes counter-propagate, so a molecule with transverse velocity v_x along the light sees $\delta + k_L v_x$ in half of the passes and $\delta - k_L v_x$ in the others ($k_L v_x / 2\pi = 3.68\,\mathrm{MHz}$ per m/s at 271.7 nm): no laser frequency compensates both. One molecule is followed through 5 cm at the mean forward velocity, with fixed v_x and fixed offset y_0 along the beam height. The figure of merit is the cleanup efficiency $\eta_\mathrm{c} = 1 - \sum_{i \in J} \rho_{ii}(T)$. Grids over (v_x, y_0) are averaged with the Gaussian distribution of a simulation of the lensed beam within the acceptance of the detection laser (Table 1.1); this *mean in acceptance* is the number quoted for the transverse distribution. Since the master equation is linear in $\rho(0)$, one trajectory per starting sublevel gives the result for any incoherent initial mixture.

Numerics. The master equation is integrated with the adaptive Dormand–Prince method [3] of the Rust backend of `centrex-tlf` (versions 0.2.5 and 0.3.0, which agree to 10^{-6} for CU1), with relative and absolute tolerances 10^{-7} and 10^{-9} (10^{-6} and 10^{-8} for the CU2 scans, which changes the CU2 optimum by 3×10^{-5}). A CU1 trajectory costs about 0.5 s in a scan, a CU2 trajectory about 40 s, because of the second beam's 200 MHz phase factor and the 17.5 MHz modulation. Scans are stored with their parameters in HDF5 files (`results/`); the scripts are in `scripts/`.

1.3 CU1

1.3.1 Scheme and on-axis optimum

At 150 V/cm the five sublevels of B, $J' = 2$, $F_1' = 5/2$, $F' = 2$, $P' = +$ form three groups: $m_F' = 0$ highest, $|m_F'| = 1$ 17.4 MHz below and $|m_F'| = 2$ a further 24.4 MHz below; the

12 sublevels of $J = 1$ span 2.4 MHz. The design study [4] started from an EOM whose sidebands reach the outer groups, but its scans of detuning against modulation frequency put the minimum of the remaining population at zero modulation for every power. A single unmodulated laser near the $|m'_F| = 1$ pair, with X/Z switching, is enough.

On the beam axis ($v_x = 0$, $y_0 = 0$), at 80 mW with a 4.8 cm high beam, the best detuning is +20.2 MHz, where CU1 removes 99.963 % of the $J = 1$ population; the removed molecules go to $J = 3$ and $J = 2$ in about equal parts. The optimum coincides with the strength-weighted centre of the transition components, +19.5 MHz, and does not depend on the power, but it is narrow: the cleanup exceeds 99.9 % only between +20.0 and +20.5 MHz, and 99 % within 2.1 MHz.

1.3.2 Transverse velocity and the set point

Because the Doppler shift alternates from pass to pass, the narrow optimum makes CU1 very sensitive to the transverse velocity. At +20.2 MHz on the axis the cleanup exceeds 99.9 % only for $|v_x| \leq 0.2$ m/s and 99 % only for $|v_x| \leq 0.4$ m/s; it falls to 97.4 % at 0.5 m/s and to 78 % at 1 m/s (Fig. 1.1). The transverse position matters less (99.5 % at $y_0 = 12$ mm).

Averaged over the velocities, the on-axis optimum is not the best detuning, because of the ground-state structure in the field. The 12 sublevels of $J = 1$ form two groups 2.4 MHz apart. The four mainly-$m_J = 0$ levels, which include the main ground state, reach the driven $|m'_F| = 1$ components at $\delta/2\pi \approx +18.5$ MHz. The eight mainly-$|m_J| = 1$ levels, into which SPB transfers the molecules (the *SPB-related* levels), reach them only at +20.7 to +20.9 MHz. This group contains the two most weakly coupled levels, $|F_1 = \frac{3}{2}, F = 2, m_F = \pm 2\rangle$ and $|F_1 = \frac{1}{2}, F = 0\rangle$ ($|M|^2 = 0.005$ and 0.003, against 0.014–0.106 for the others). At +19.0 MHz the laser is 1.7–1.9 MHz below this group for a molecule at rest, and the two weak levels keep about 9 % of their population; a molecule with $|v_x| \approx 0.5$ m/s, however, sees the laser 1.84 MHz higher in half of the passes and is emptied completely. Since most molecules move transversely, +19.0 MHz is better on average (Table 1.2), and it was taken as the set point. A second maximum lies at +22.0 to +22.5 MHz. A 1.6 MHz phase EOM that broadens the laser over the Doppler shifts gains less than 1 % and was not adopted.

Table 1.2: CU1 model predictions: cleanup on the axis and mean in the detection acceptance with the lensed-beam distribution ($\sigma_{v_x} = 0.65$ m/s).

power, beam height, detuning	initial state	on axis	mean in acceptance
80 mW, 4.8 cm, +20.2 MHz	SPB-related levels	99.96 %	95.5 %
80 mW, 4.8 cm, +19.0 MHz	SPB-related levels	96.4 %	97.1 %
80 mW, 3 cm, +19.0 MHz	SPB-related levels	99.0 %	98.7 %
50 mW, 2 cm, +19.0 MHz	SPB-related levels	98.8 %	96.5 %
50 mW, 2 cm, +19.0 MHz	AP1 triplet / singlet	99.92 % / 99.93 %	98.65 % / 99.68 %

The test of Sec. 1.3.3 used 50 mW in a 2 cm beam (Fig. 1.2). The best detuning is again +19.0 MHz, but the mean in acceptance drops from 98.7 to 96.5 %: the peak intensity is similar to that of 80 mW in 3 cm, but at $y_0 = 8.5$ mm the 2 cm beam has only 24 % of its peak intensity, against 53 %. The test molecules come from AP1, which populates the four mainly-

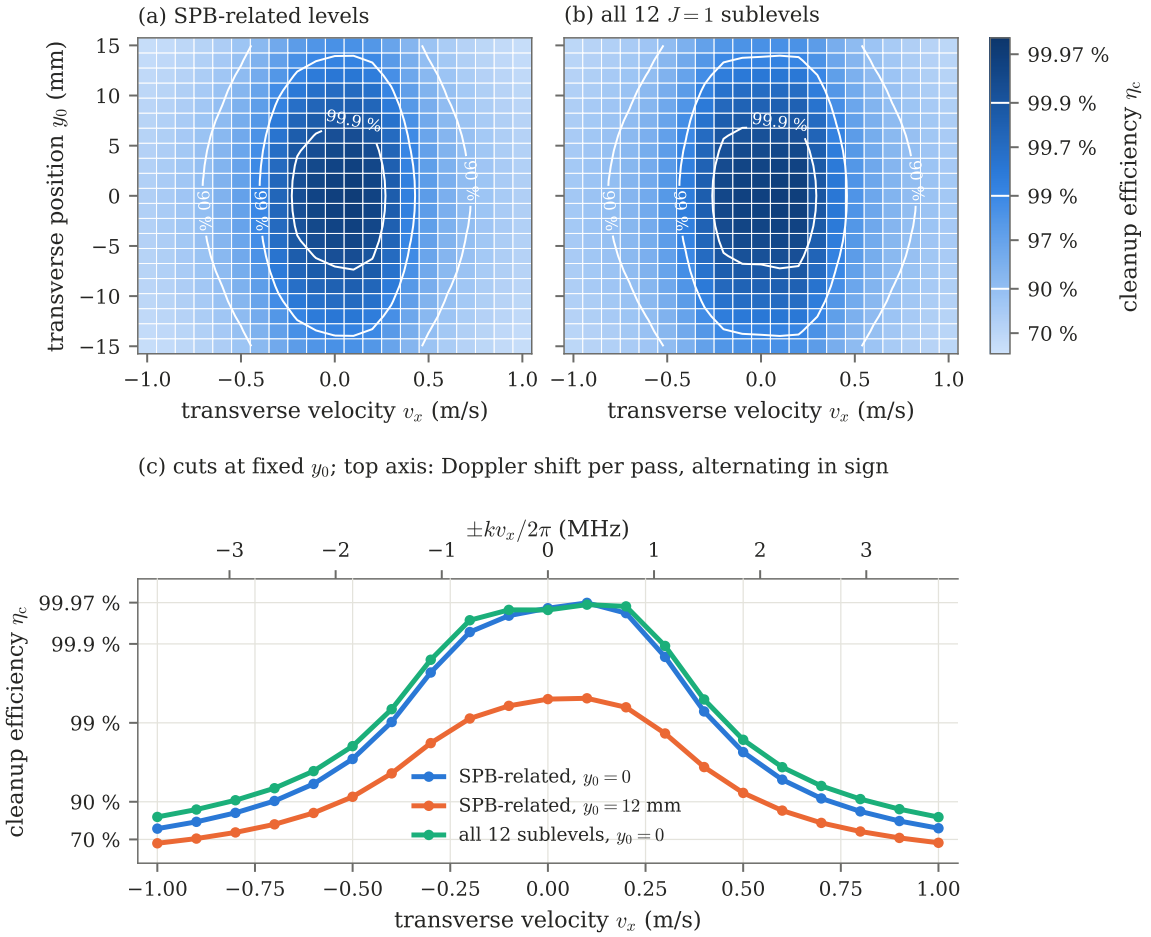


Figure 1.1: CU1 cleanup versus transverse velocity v_x and position y_0 at the on-axis optimum, +20.2 MHz (80 mW, 4.8 cm beam): (a) SPB-related start, (b) uniform 12-level start, logarithmic colour scale with contours at 90, 99 and 99.9 %; (c) cuts at fixed y_0, with the Doppler shift per pass on the top axis.

$m_J = 0$ levels (Sec. 1.3.3); they are close to resonance at +19.0 MHz and are cleaned better. A tilt θ of the laser in the plane of the molecular beam adds $v_z\theta$ to the effective transverse velocity, with the same alternating sign (0.18 m/s per mrad), which re-tuning cannot remove. The mean in acceptance then drops by 0.4–0.7 % at 1 mrad and by 4–5 % at 3 mrad: the laser must be perpendicular to the molecular beam within about 1 mrad.

1.3.3 Test on the SPB setup

Setup and method. CU1 was tested on the SPB setup between 2026-10-16 and 2026-10-21 [5]. Rotational cooling (RC) and the first adiabatic passage of SPA (AP1) prepare the $J = 1$ molecules. AP1 transfers all four $J = 0$ hyperfine sublevels to the four mainly-$m_J = 0$ levels of $J = 1$, the triplet $|F_1 = \frac{3}{2}, F = 2, m_F = 0, \pm 1\rangle$ and the singlet $|F_1 = \frac{3}{2}, F = 1, m_F = 0\rangle$ [6]. RC on P(2) $F_1' = 3/2$, $F' = 2$ enhances $J = 0$, $F = 1$, so the triplet dominates. The lens (±24.7 kV) hardly acts on these molecules. CU1 is 2.5 m from the cell, at 156 V/cm, with 50 mW (about 45 mW entering the multipass) in a 2 cm beam. The detection laser, 3.15 m

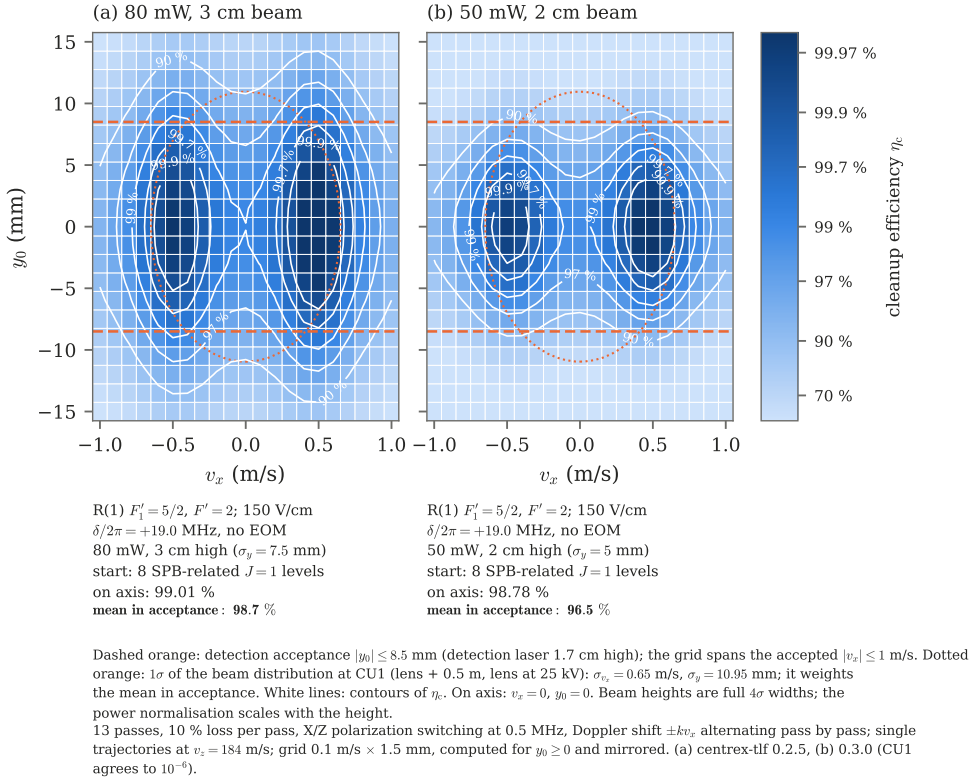


Figure 1.2: CU1 cleanup versus transverse velocity v_x and position y_0 at the set point +19.0 MHz, SPB-related start: (a) 80 mW, 3 cm beam; (b) 50 mW, 2 cm beam of the test. Dashed: detection acceptance; dotted ellipse: 1σ of the beam distribution at CU1. The maxima near $|v_x| = 0.5$ m/s are where the Doppler shift brings the SPB-related levels into resonance in half of the passes.

from the cell, drives R(1) $F_1' = 5/2$, $F' = 3$. This line couples only the five $F_1 = \frac{3}{2}$, $F = 2$ sublevels and has no dark state, so as long as every $F = 2$ molecule is pumped out, the ratio of two signals is the ratio of the $F = 2$ populations. It propagates parallel to CU1 and accepts $|v_x|$ up to about 1.2–1.7 m/s; in the final experiment the detection, 8 m downstream behind a 2 cm aperture, accepts only $|v_x| \leq 0.22$ m/s. A shutter in the CU1 beam alternates between open and closed on consecutive ablation shots, in bursts of 10 open and 10 closed shots on one target spot, so that source and absorption drifts cancel in the remaining fraction $r = A_{\text{open}}/A_{\text{closed}}$, the ratio of the fitted fluorescence amplitudes. The efficiency is $\eta_c = 1 - r$, with uncertainties from a bootstrap over bursts.

Progress and final result. The remaining fraction fell from 0.099 to 0.026 within two days (Fig. 1.4). The main steps were the alignment of the multipass: a cross alignment laser put the CU1 beam, the detection beam and the chamber centre at the same height; the number of passes was raised from 13 to about 17; and the reflected beam was overlapped with the incoming beam, so that the laser is perpendicular to the molecular beam. RC was moved from $F' = 1$ to $F' = 2$: with $F' = 1$ the singlet dominates, and CU1 pumps part of it into $F = 2$, where R(1) sees it, while R(1) cannot see the singlet with CU1 closed. The set point was moved to the measured optimum, which lies 4–6 MHz (UV) below the predicted

+19 to +20 MHz. The final run, 2000 shots with alignment C, gave $r = 0.026$ $[0.024, 0.028]$ (Fig. 1.3), i.e.

$$\eta_c = (97.4 \pm 0.2)\,\%.$$

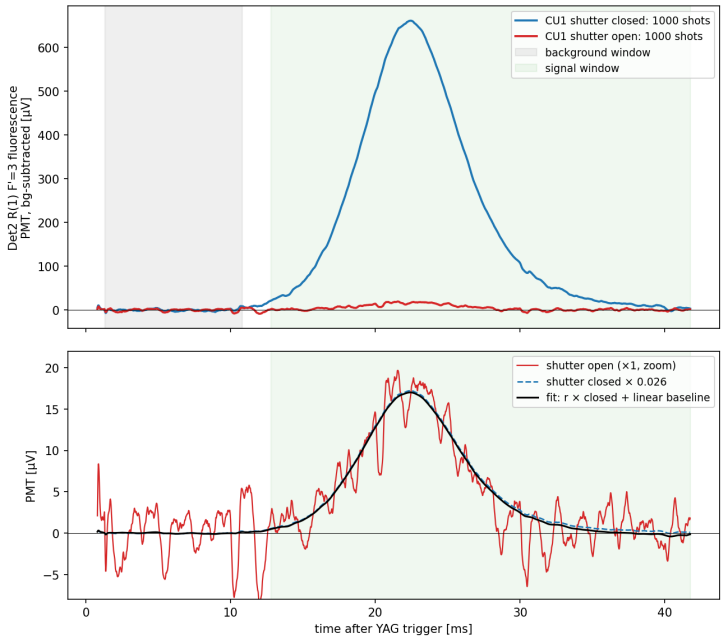


Figure 1.3: CU1 test, final run (10-19, 2000 shots): mean detection traces with the CU1 shutter closed and open (top), and the open-shutter trace enlarged with the fit $r \times$ closed + linear baseline (bottom). Grey: background window; green: signal window.

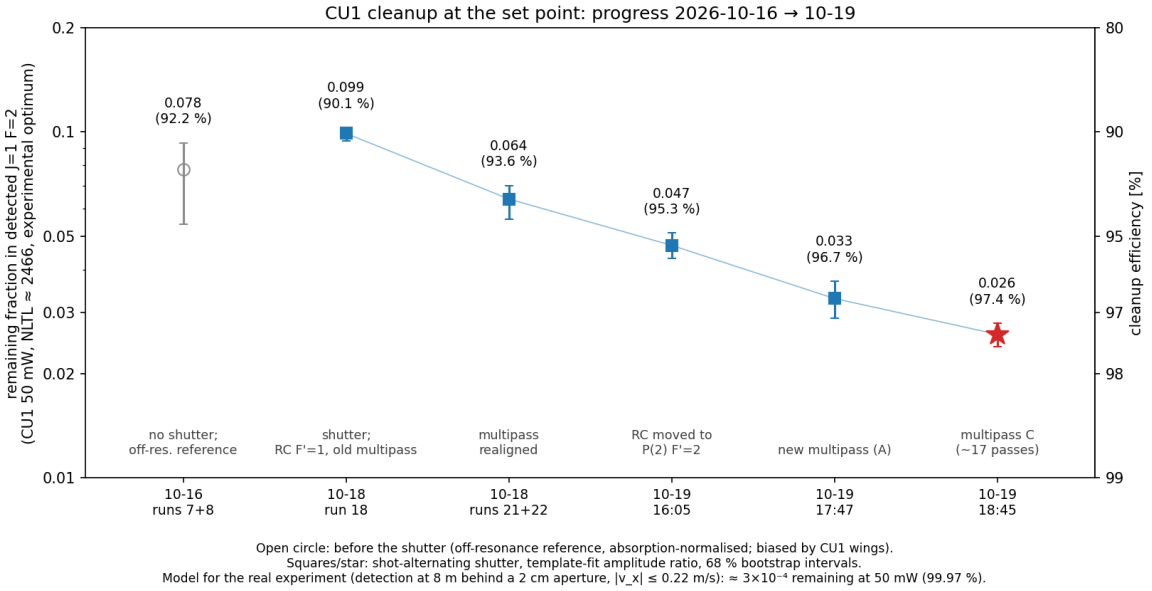


Figure 1.4: CU1 test: remaining fraction of the detected $J = 1$, $F = 2$ population at 50 mW from 10-16 to 10-19, with the change made before each point. Open circle: before the shutter, with an off-resonance reference; squares and star: shutter method, 68 % bootstrap intervals.

Power dependence and transverse velocity. The remaining fraction falls only slowly with the CU1 power P: a fit of $r = \exp(-kP^a)$ from 13 to 50 mW gives $a = 0.45 \pm 0.03$, whereas a homogeneous ensemble gives $a = 1$. This points to a wide distribution of Doppler shifts. A scan of the detection laser across R(1) $F' = 3$ with CU1 off measures the v_x distribution of the detected molecules directly (Fig. 1.5): $\sigma_{v_x} = (1.28 \pm 0.03)$ m/s, the same at 100 and 400 µW, and close to a simulation of the $J = 1$ beam through the lens (1.37 m/s). It is twice the value assumed in Sec. 1.3.2, which was that of the lensed $J = 2$ beam.

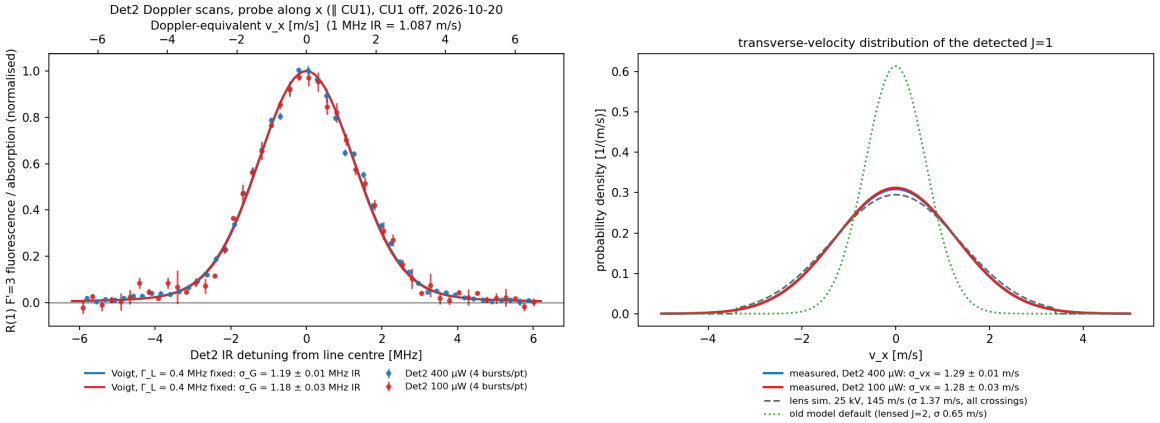


Figure 1.5: Doppler scan of R(1) $F' = 3$ with the detection laser parallel to CU1, CU1 off. Left: signal versus IR detuning with Voigt fits (natural Lorentzian width fixed); top axis: equivalent v_x. Right: the resulting v_x distributions, with the lens simulation for $J = 1$ and the distribution assumed before the test.

Comparison with the model. With the measured distribution ($|v_x| \leq 4$ m/s), $v_z = 145$ m/s and the AP1 triplet start, the model shows that the residual comes from the molecules with $|v_x| > 1$ m/s (Fig. 1.6a): at 50 mW almost nothing is left below 0.6 m/s, 0.026 at 1 m/s and 0.20 at 2 m/s. With 17 passes and 3.5 % loss per pass, the one adjusted parameter, the model reproduces the power dependence measured at the optimum within 5 % at 30–50 mW, 0.031 against 0.032 at 50 mW (Fig. 1.7). With the earlier distribution the power would have had to be scaled down by a factor of four. Inside the acceptance of the final experiment (green band in Fig. 1.6a) the model predicts a remaining fraction of 2×10^{-5} (17 passes) to 3×10^{-4} (13 passes, 10 % loss) at 50 mW: there CU1 will remove more than 99.9 % of the $J = 1$ population.

Q(1) detection. Q(1) $F_1' = 1/2$, $F' = 1$ couples all 12 sublevels of $J = 1$, but with 12 ground and 3 excited sublevels the molecules are pumped into dark states during the detection, so the Q(1) ratio is not the remaining fraction of the whole $J = 1$ level. With RC on $F' = 1$ it gave (95.1 ± 0.4) %, against about 97 % with R(1) on the same day, likely because some sublevels that Q(1) sees are cleaned less well. This was not pursued further.

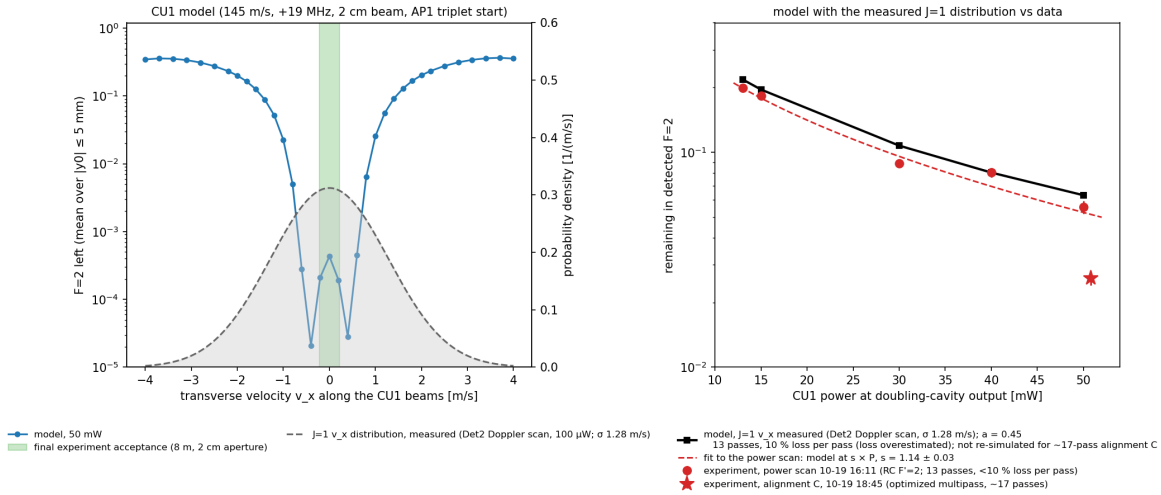


Figure 1.6: CU1 model with the measured v_x distribution. (a) Remaining $F = 2$ population versus v_x at 50 mW (145 m/s, +19 MHz, 2 cm beam, AP1 triplet start), with the measured distribution (grey) and the acceptance of the final experiment (green). (b) The 13-pass model with this distribution against the power scan of 10-19, and its fit with a power scale $s = 1.14 \pm 0.03$.

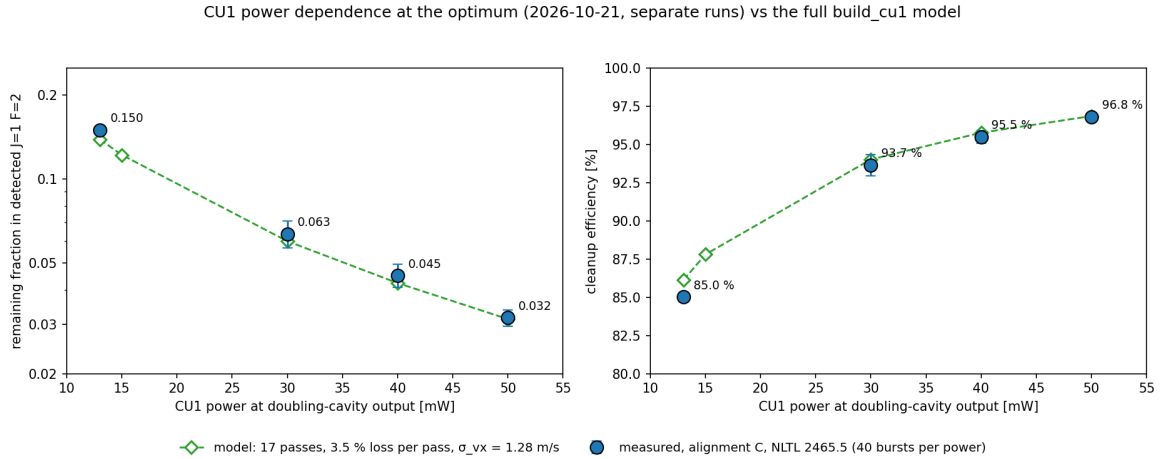


Figure 1.7: CU1 power dependence at the optimum (10-21, alignment C, 40 bursts per power) against the model with 17 passes, 3.5 % loss per pass and $\sigma_{v_x} = 1.28$ m/s: remaining fraction (left) and efficiency (right).

1.4 CU2

1.4.1 Scheme

At the original field of 250 V/cm the sublevels of B, $J' = 3$, $F_1' = 7/2$, $F' = 3$, $P' = -$ are split by about 18 MHz per step in $|m_F'|$, and the 20 sublevels of $J = 2$ span 1.4 MHz. The design study [4] reached the scheme in three steps. One laser on $F' = 3$ with a resonant EOM (about 17 MHz, $\beta \approx 1.5$, 100 mW) cleans only about 98 %: $|J = 2, F_1 = \frac{3}{2}, F = 1, m_F = 0\rangle$ reaches $F' = 3$ only through its admixture of $F = 3$, and only the $m_F' = \pm 1$ components, through the sidebands, so it keeps about a fifth of its population. A second beam on $F' = 4$, about 200 MHz higher and made from the same laser by an AOM, removes this bottleneck

with the same total power. Finally, one EOM placed before the AOM modulates both beams; this costs little (about 99.6 % at 60 + 18 mW, against 99.95 % with an EOM on each beam) and is much simpler. In the adopted layout the undiffracted and first-order outputs of the AOM are recombined with orthogonal polarizations and pass a common polarization EOM, so the two beams switch between X and Z in antiphase.

The AOM couples the two detunings. With δ_k measured from the main resonance of beam k, the AOM frequency is the frequency difference of the beams,

$$f_{\mathrm{AOM}} = \Delta_{\mathrm{main}}(E) + \frac{\delta_1 - \delta_0}{2\pi}, \tag{1.3}$$

where Δ_{main} is the separation of the two main resonances, 211.83 MHz at 200 V/cm and 219.93 MHz at 250 V/cm. The laser frequency sets δ_0 and the AOM frequency $\delta_1 - \delta_0$; the AOM range 200 ± 15 MHz limits $(\delta_1 - \delta_0)/2\pi$ to $-26.83 \ldots + 3.17$ MHz at 200 V/cm. The components driven by each beam spread over about 87 MHz ($F' = 3$: 0 to +87 MHz from its main resonance; $F' = 4$: −31 to +56 MHz), which is why both beams are phase-modulated.

1.4.2 Optimisation at 200 V/cm

CU2 now operates at about 200 V/cm. The detunings chosen at 250 V/cm, $(\delta_0, \delta_1)/2\pi = (+18.5, -6.5\,\mathrm{MHz})$, which gave 99.99 % there, give only 92.9 % at 200 V/cm: Δ_{main} has moved by 8.1 MHz, and the component positions within each manifold have moved as well. A scan of the laser frequency at the AOM frequency that keeps the former detuning difference (186.83 MHz) reaches 99.33 % at best; laser and AOM frequency must be optimised together. The two-dimensional scan uses the experimental coordinates $(\delta_0, f_{\mathrm{AOM}})$: a coarse 2 MHz grid over $\delta_0/2\pi = +5 \ldots + 23$ MHz, about one EOM period, and the full AOM range (160 points, Fig. 1.8); 0.5 MHz grids around its three best local maxima (219 points); and a check of the best points with the tighter tolerances. Only 15 of the 160 coarse points exceed 99 %. The fine grids give the optima of Table 1.4: R2 with 99.85 % at (+11.5 MHz, 191.0 MHz) and, nearly equivalent, R1 with 99.80 % at (+11.5 MHz, 208.5 MHz), both well inside the AOM range.

Both optima are 0.5–1.5 MHz wide at the 99.7 % level and 2–3 MHz wide at 99 % (Table 1.3). They are elongated along lines of constant δ_1 (Fig. 1.9): δ_1 must be held to about ±0.5 MHz, whereas δ_0 tolerates ±1.5 MHz if δ_1 is kept fixed. A drift of the laser frequency, or of the field, shifts both detunings together and is therefore limited by the tighter δ_1 tolerance; the field variation of about 2 V/cm (2 MHz) across the cleanup region estimated in the design study is comparable to the 99 % width.

1.4.3 Two optima and the bottleneck

R1 and R2 differ by 17.5 MHz in f_{AOM}, exactly one EOM period, with beam 0 unchanged. The comb of beam 1, at $\delta_1/2\pi + n \times 17.5$ MHz, therefore occupies the same frequencies in both settings, and only the order n of each line differs by one (Fig. 1.10). At $\beta = 1.5$ the carrier carries 26 % of the power and each first-order sideband 31 %. The two lines that

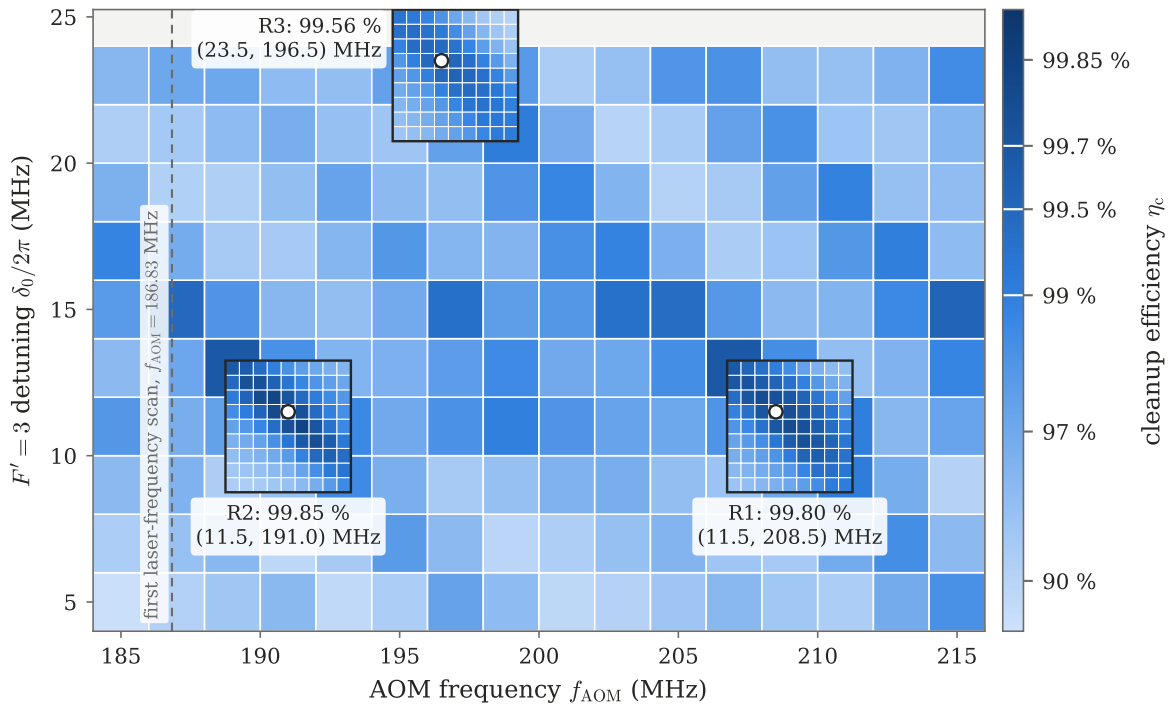


Figure 1.8: CU2 cleanup at 200 V/cm versus the AOM frequency (full range) and the $F' = 3$ detuning δ_0, i.e. the laser frequency, on axis, 2.4 cm beam. Colour: remaining $J = 2$ population on a logarithmic scale, labelled in cleanup efficiency. Large cells: coarse 2 MHz grid; outlined boxes: 0.5 MHz grids R1–R3 with their best points (circles). Dashed: the AOM frequency of the first laser-frequency scan.

Table 1.3: CU2 at 200 V/cm: ranges (on the 0.5 MHz grid) within which the cleanup exceeds the given level, through the best points of R1 and R2.

optimum	direction	$> 99.7\,\%$	$> 99.5\,\%$	$> 99\,\%$
R2	$\delta_0/2\pi$ at $f_{AOM} = 191.0$ MHz	11.0–12.0 MHz	11.0–12.0 MHz	10.5–12.5 MHz
R2	f_{AOM} at $\delta_0/2\pi = 11.5$ MHz	190.5–191.5 MHz	190.0–191.5 MHz	189.5–192.0 MHz
R1	$\delta_0/2\pi$ at $f_{AOM} = 208.5$ MHz	11.5–12.0 MHz	11.0–12.5 MHz	10.5–13.0 MHz
R1	f_{AOM} at $\delta_0/2\pi = 11.5$ MHz	208.0–209.5 MHz	208.0–210.0 MHz	207.5–210.5 MHz

Table 1.4: CU2 optima at 200 V/cm (on axis, 2.4 cm beam, uniform start), with the share of the remaining population in the stretched sublevels $|F_1 = \frac{5}{2}, F = 3, m_F = \pm 3\rangle$. Efficiencies of R1 and R2 with tight tolerances.

optimum	$\delta_0/2\pi$	f_{AOM}	$\delta_1/2\pi$	η_c	stretched share
R1	11.5 MHz	208.5 MHz	+8.17 MHz	99.82 %	74 %
R2	11.5 MHz	191.0 MHz	−9.33 MHz	99.85 %	56 %
R3	23.5 MHz	196.5 MHz	+8.17 MHz	99.56 %	–

matter most sit on the strongest $F' = 4$ component, $m'_F = 0$ at +8.65 MHz, and on $m'_F = \pm 2$ at −10.32 MHz; these are 19.0 MHz apart, close to the comb spacing, and receive nearly the same power in both settings (Table 1.5). Beam 0 is the same in both: its first-order sideband sits on the $F' = 3$, $m'_F = \pm 1$ components and its carrier 3 MHz below $m'_F = \pm 2$. Within the AOM range only these two copies of the optimum exist.

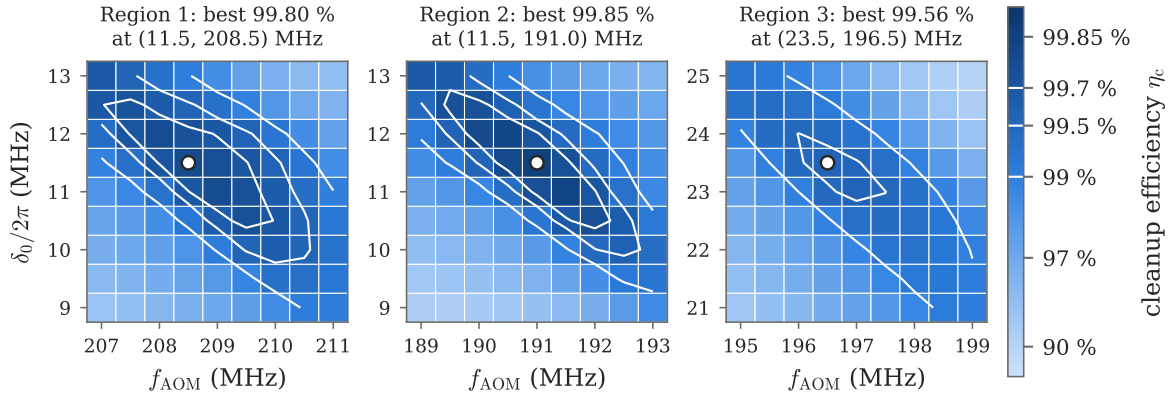


Figure 1.9: CU2 at 200 V/cm: 0.5 MHz grids around the optima R1–R3, same colour scale as Fig. 1.8. White contours: 99, 99.5 and 99.7 %; circles: best points. The high-efficiency regions run along lines of constant δ_1 (slope -1).

Table 1.5: CU2 at 200 V/cm: comb lines of beam 1 near the $F' = 4$ sublevels, on the δ_1 axis, with their order n and power fraction $J_n(1.5)^2$ at R1 and R2.

line (MHz)	nearest $F' = 4$ sublevel(s) (MHz)	R1: n, power	R2: n, power
+8.17	$m'_F = 0$, $P' = -$, at +8.65 (strongest)	0, 26 %	+1, 31 %
−9.33	$m'_F = \pm 2$, $P' = -$, at −10.32	−1, 31 %	0, 26 %
+25.67	$m'_F = \pm 1$, $P' = +$, at +22.50	+1, 31 %	+2, 5 %
−26.83	$m'_F = \pm 3$ at −20.83, ± 4 at −31.41, $P' = -$	−2, 5 %	−1, 31 %
+43.17	$m'_F = \pm 3$, $P' = +$, at +44.43	+2, 5 %	+3, 0.4 %

At the optima the first pass removes about 75 % of the $J = 2$ population, and each later pass multiplies what is left by 0.5–0.8, following the alternating pass powers (Fig. 1.11). The removed molecules go to $J = 3$ (53–54 %), $J = 4$ (44–45 %) and $J = 5$ (1.7 %), with 3×10^{-5} in $J = 1$, after 1.5 photons per molecule. The residual is dominated by the stretched sublevels $|F_1 = \frac{5}{2}, F = 3, m_F = \pm 3\rangle$ (Fig. 1.12, Table 1.4). They are pumped almost only by the weaker $F' = 4$ beam, through $m'_F = \pm 3$ and ± 4 at about -20 and -31 MHz, far from the two main lines. R2 has a first-order sideband 4.6 MHz from $m'_F = \pm 4$, R1 only a second-order one, so R2 pumps the stretched sublevels 15 % faster and ends with less population left; R1 in turn pumps the $F_1 = \frac{3}{2}$, $F = 2$ sublevels faster.

1.4.4 Transverse motion at the present beam height

In the experiment the CU2 beams are now about 3 cm high at the same 80 + 24 mW, 0.8 times the peak intensity of the 2.4 cm beam used for the optimisation. The two beams share the multipass, so the Doppler shift is common to both. Table 1.6 and Fig. 1.13 give R1 and R2 over the transverse velocity and position, for all 20 sublevels and for the four SPB source levels ($J = 2$, $m_J = 0$, weighted 2/3 for the singlet and 1/9 for each triplet level).

- **The transverse position dominates.** The intensity falls to 73, 49 and 28 % of its axial value at $y_0 = 6$, 9 and 12 mm, and from the uniform start the cleanup falls from

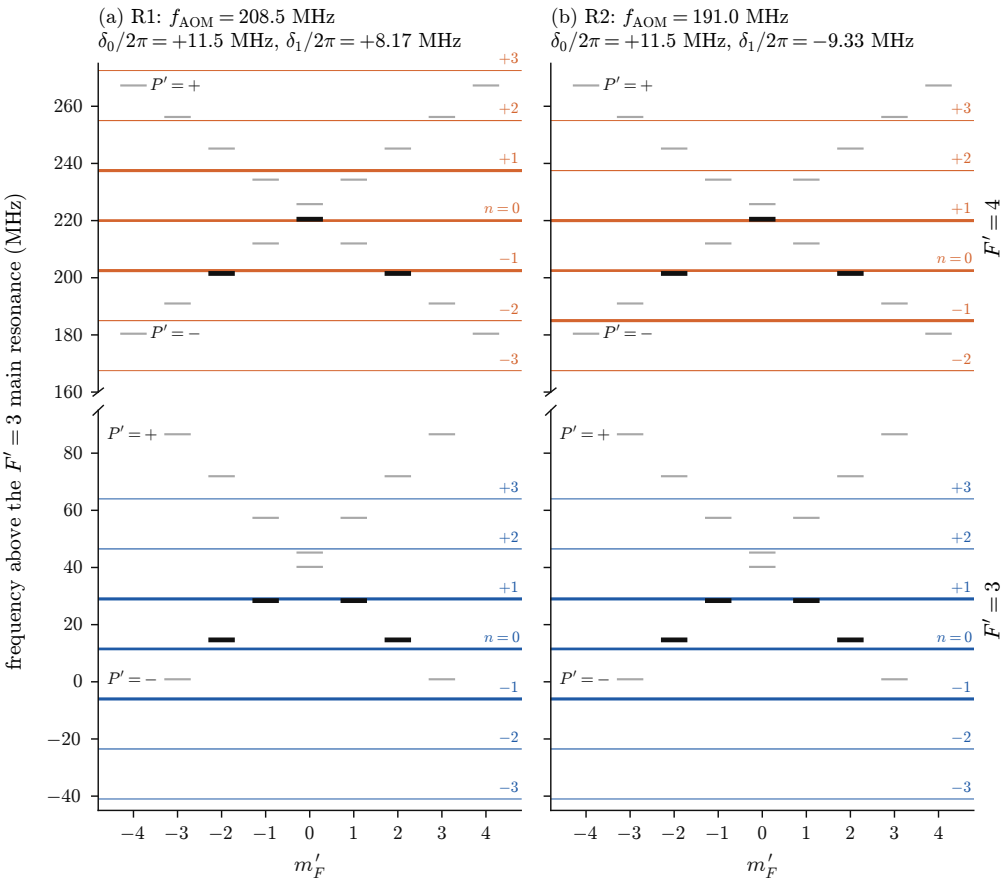


Figure 1.10: CU2 at 200 V/cm: excited sublevels of B, $J' = 3$, $F'_1 = 7/2$ addressed from X, $J = 2$, and the laser combs at (a) R1 and (b) R2. Frequency above the $F' = 3$ main resonance, axis broken between $F' = 3$ (bottom) and $F' = 4$ (top). Bold: sublevels driven by the light. Coloured lines: carrier and sidebands of beam 0 (blue) and beam 1 (orange), line width proportional to the power fraction. The beam-1 lines lie at the same frequencies in both settings; only their orders differ by one.

99.5 % on the axis to 96.0 % at 9 mm.

- **The velocity matters much less than for CU1**: 97.8–98.4 % at $|v_x| = 0.4$ m/s and 94.7–95.5 % at 1 m/s on the axis. Each beam carries a comb of five significant lines on 32 sublevels, so a shift of a few MHz moves some lines off their components while others come closer.
- **R1 is the more robust setting** for both starts and is recommended: within the acceptance it removes 96.7 % of a uniformly populated $J = 2$ level and 99.6 % of the SPB source levels.

For the $J = 2$ molecules that SPB leaves behind, which the lens has focused, the lensed-beam distribution is the right input; molecules that CU1 pumps into $J = 2$ have the wider $J = 1$ distribution of Sec. 1.3.3. The CU2 results were computed with `centrex-tlf` 0.2.5. Version 0.2.8 restored the Stark mixing with lower J in the X Hamiltonian, which moves $J = 2$ line positions by up to about 1 MHz, so the optimum must be re-run, together with a re-optimisation for the 3 cm beam, before it is relied on at the sub-MHz level.

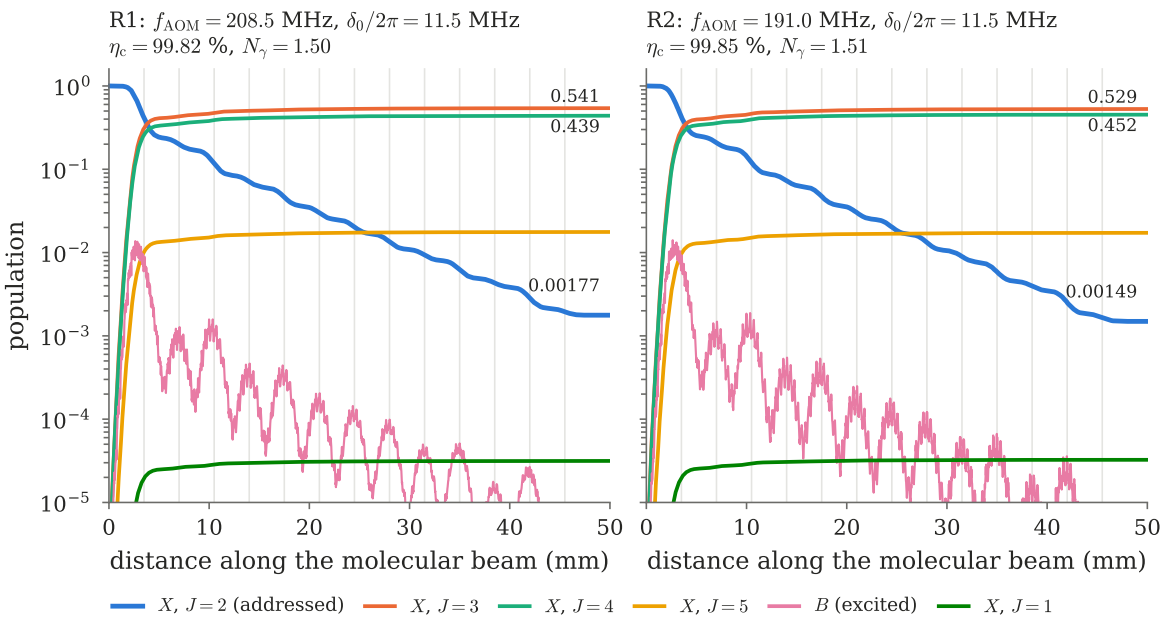


Figure 1.11: CU2 at 200 V/cm: populations of the rotational levels along the interaction region at R1 and R2, logarithmic scale. Grey lines: pass centres. The excited-state population follows the 0.5 MHz polarization switching and the 17.5 MHz modulation.

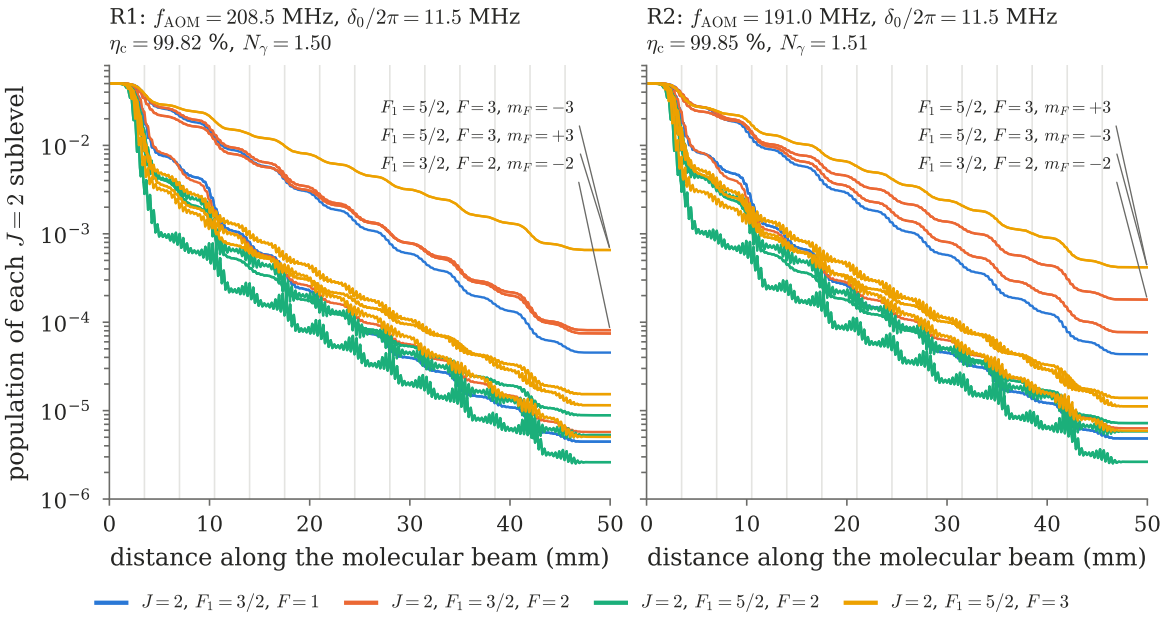


Figure 1.12: CU2 at 200 V/cm: populations of the 20 $J = 2$ sublevels along the interaction region at R1 and R2, coloured by hyperfine level (F_1, F); the three sublevels with the most population left are labelled.

Table 1.6: CU2 at 200 V/cm with the 3 cm beam: cleanup on the axis, averaged over v_x at $y_0 = 0$, and mean in the detection acceptance (coarse grid, lensed-beam weights). The recommended setting is in bold.

setting	initial state	on axis	v_x-averaged, $y_0 = 0$	mean in acceptance
R2, 191.0 MHz	all 20 sublevels	99.59 %	97.72 %	96.10 %
R1, 208.5 MHz	all 20 sublevels	99.52 %	98.16 %	96.66 %
R2, 191.0 MHz	SPB source levels	99.975 %	99.67 %	99.35 %
R1, 208.5 MHz	SPB source levels	99.974 %	99.81 %	99.57 %

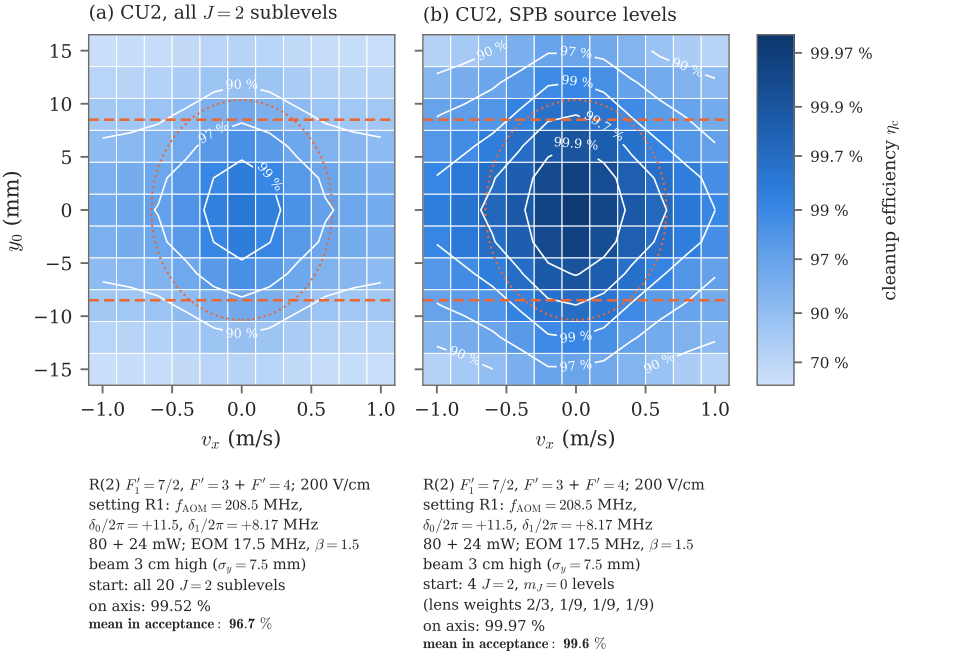


Figure 1.13: CU2 at R1 ($f_{AOM} = 208.5$ MHz, $\delta_0/2\pi = +11.5$ MHz), 3 cm beam, 80 + 24 mW: cleanup versus transverse velocity and position from (a) all 20 $J = 2$ sublevels and (b) the four SPB source levels. Dashed: detection acceptance; dotted ellipse: 1σ of the beam distribution at CU2.

1.5 Conclusions and outlook

CU1 uses one unmodulated laser on R(1) $F_1' = 5/2$, $F' = 2$ at 150 V/cm. On the axis it removes 99.96 % of the $J = 1$ population, but its optimum is only 0.6 MHz wide, and the Doppler shift, alternating from pass to pass, makes it sensitive to the transverse velocity. The set point +19.0 MHz uses the Doppler shift to reach the SPB-related levels. In the test on the SPB setup CU1 removed (97.4 ± 0.2) % of the detected $F = 2$ molecules at 50 mW. The slow power dependence and the measured transverse velocities, $\sigma_{v_x} = 1.28$ m/s, show that the residual comes from $|v_x| > 1$ m/s, which the test detection accepts. With the measured distribution the model reproduces the data, and it predicts more than 99.9 % for the acceptance of the final experiment.

Table 1.7: Main results.

CU1, on axis, 80 mW, +20.2 MHz; width of the optimum at 99.9 %	99.96 %; 0.6 MHz
CU1 set point; mean in acceptance at 80 mW, 3 cm (model, $\sigma_{v_x} = 0.65$ m/s)	+19.0 MHz; 98.7 %
CU1 test, best cleanup (R(1) $F' = 3$ detection, 50 mW, 2 cm)	(97.4 ± 0.2) %
CU1 test, power exponent a; σ_{v_x} of the detected $J = 1$ molecules	0.45 ± 0.03; (1.28 ± 0.03) m/s
CU1, remaining at 50 mW: model with the measured σ_{v_x} / data	0.031 / 0.032
CU1, predicted for the final experiment ($\lvert v_x \rvert \leq 0.22$ m/s)	> 99.9 %
CU2 optima R2 / R1 at 200 V/cm (on axis, 2.4 cm)	99.85 % / 99.80 %
CU2, width at 99 % along laser / AOM frequency	2 / 2.5 MHz
CU2, R1 with the 3 cm beam, mean in acceptance: all sublevels / SPB source levels	96.7 % / 99.6 %

CU2 uses two beams from one laser, on $F' = 3$ and $F' = 4$ of R(2) $F_1' = 7/2$, coupled by an AOM and phase-modulated by one EOM. At 200 V/cm laser and AOM frequency had to be optimised together; the optima, 99.85 % at $f_{\mathrm{AOM}} = 191.0$ MHz and 99.80 % at 208.5 MHz, lie one EOM period apart and are 2–3 MHz wide at 99 %. With the present 3 cm beam R1 ($f_{\mathrm{AOM}} = 208.5$ MHz) is the more robust setting: 96.7 % of a uniformly populated $J = 2$ level and 99.6 % of the SPB source levels within the acceptance, limited by the beam edges and by the stretched sublevels.

Next steps:

1. CU1: test the velocity interpretation with a narrower detection acceptance; find the origin of the 4–6 MHz offset of the measured optimum; recompute the set point with the measured inputs ($\sigma_{v_x} = 1.28$ m/s, 145–155 m/s, 17 passes).
2. CU2: re-run with `centrex-tlf` 0.2.8 or later, re-optimise laser and AOM frequency and the power split for the 3 cm beam, check the field uniformity at 200 V/cm, and test CU2 on the setup.
3. Both: take the transverse distribution of the molecules that reach the detection laser from the beam simulation, separately for the $J = 1$ molecules, which the lens does not focus.

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